

A USB Type-C Multi-Port Switching Architecture for High-Speed Peripheral and Power Sharing Applications

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Abstract—USB Type-C interfaces have become increasingly important in modern electronic systems due to their support for high-speed data transfer, reversible connectivity, and power delivery capabilities. This paper presents the architecture and implementation of a USB Type-C multi-port switching system designed for high-speed peripheral and power sharing applications. The proposed system enables switching between four USB Type-C input ports and a single USB Type-C output port while maintaining USB 3.0 data communication, VBUS power routing, and CC signal transparency between connected devices.

The proposed design supports USB-based peripherals such as storage devices, keyboards, printers, and charging applications through transparent USB Type-C connectivity. Design limitations associated with DisplayPort Alternate Mode and high-speed video transmission are also discussed. The work demonstrates a practical USB Type-C switching architecture suitable for peripheral sharing and power routing applications.

Index Terms—USB Type-C, USB 3.0, High-Speed Data Routing, Power Sharing, CC Signal Routing, USB Switching Architecture

I. INTRODUCTION

USB Type-C has emerged as a widely adopted interface standard for modern electronic devices due to its support for reversible connectivity, high-speed data transfer, and power delivery capabilities. The increasing use of USB-based peripherals such as external storage devices, keyboards, printers, docking stations, and charging accessories has created a growing demand for efficient peripheral sharing and power routing systems. Conventional USB switching solutions are often limited in terms of port flexibility, high-speed signal handling, and compatibility with modern USB Type-C interfaces.

High-speed USB 3.0 systems introduce additional design challenges associated with differential signal routing, signal integrity, orientation handling, and compact PCB implementation. In USB Type-C interfaces, reversible plug orientation requires dynamic routing of high-speed TX/RX lanes to ensure correct communication between connected devices. In addition, transparent routing of Configuration Channel (CC) signals is necessary for proper USB Type-C connectivity and power negotiation between endpoint devices.

This paper presents a USB Type-C multi-port switching architecture designed for high-speed peripheral and power

sharing applications. The proposed system implements a four-input and single-output USB Type-C switching topology capable of routing USB 3.0 data signals, VBUS power, and CC communication paths between connected devices. The architecture integrates dedicated hardware blocks for orientation detection, high-speed data multiplexing, power-path switching, and CC signal routing. A rotary-switch based user control mechanism is implemented to select the active USB Type-C input channel.

The PCB layout of the proposed system is designed within an 80 mm × 80 mm form factor while considering practical high-speed design constraints such as differential pair symmetry, compact component placement, and routing complexity associated with USB 3.0 interfaces. The proposed design supports USB peripheral sharing and power delivery applications while maintaining transparent USB Type-C communication between connected devices. Limitations associated with DisplayPort Alternate Mode and high-speed video transmission are also discussed as part of the system-level analysis.

II. SYSTEM ARCHITECTURE

A. Overall System Architecture

The proposed system implements a multi-port USB Type-C switching architecture consisting of four USB Type-C input ports and a single USB Type-C output port. The architecture is designed to route high-speed USB 3.0 data signals, VBUS power, and Configuration Channel (CC) communication paths between the selected input device and the output port. The overall system is organized into three primary functional blocks: the high-speed data routing path, the power routing path, and the CC signal routing path.

In high-speed USB Type-C systems, all three paths must operate together to maintain transparent communication between connected endpoint devices. Routing only USB data signals without proper CC routing may prevent correct USB Type-C negotiation and power configuration. Similarly, improper power-path handling may lead to unstable device operation or back-powering conditions. Therefore, the proposed architecture switches the USB data path, VBUS path, and CC

communication path simultaneously based on the selected input channel.

A rotary switch-based user control mechanism is used to select the active USB Type-C input port. The rotary switch generates SEL_A and SEL_B control signals which are distributed to the high-speed data multiplexer, power multiplexer, and CC multiplexer blocks. This synchronized control mechanism ensures that all switching subsystems remain aligned to the same active USB Type-C channel during operation.

The proposed architecture follows a modular design approach to simplify signal routing and improve system scalability. Each USB Type-C input channel contains dedicated protection and orientation-handling circuitry before entering the centralized switching architecture. The modular implementation also simplifies PCB floor-planning and enables easier management of high-speed differential routing constraints associated with USB 3.0 interfaces.

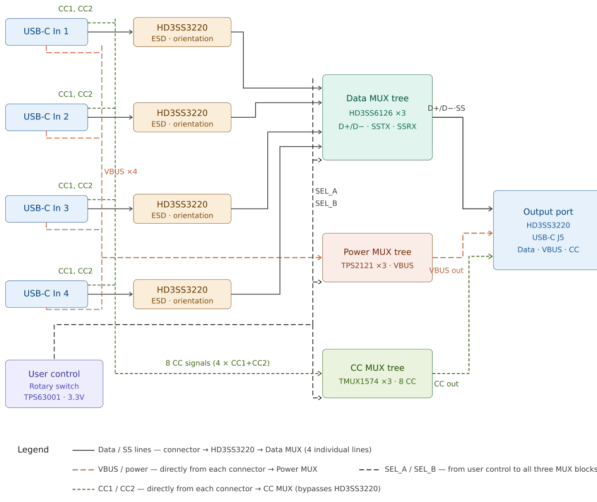


Fig. 1. Overall architecture of the proposed USB Type-C multi-port switching system.

B. USB Type-C Input Channel Architecture

Each USB Type-C input channel in the proposed system consists of a USB Type-C receptacle interface implemented using the CNC Tech C-ARA1-AK511-1 connector [5], ESD protection circuitry, and orientation detection hardware. The input-stage architecture is designed to provide reliable USB 3.0 high-speed operation while protecting sensitive switching components from electrostatic discharge events and maintaining correct signal routing behavior for reversible USB Type-C connectivity.

The USB Type-C receptacle contains multiple categories of signals including USB 2.0 D+/D- lines, high-speed SuperSpeed differential TX/RX pairs, Configuration Channel (CC) signals, VBUS power, SBU lines, and ground connections. Since USB Type-C connectors are externally accessible interfaces, they are highly susceptible to electrostatic discharge

events generated during cable insertion and removal. To improve system robustness, dedicated ESD protection devices are integrated at the input stage before signals enter the switching architecture.

For high-speed USB 3.0 differential signal protection, the PUSB3TB6 ESD protection IC is used. The device protects the SuperSpeed TX/RX differential pairs and USB 2.0 D+/D- lines from transient voltage spikes while introducing minimal parasitic capacitance to the high-speed communication path. Low parasitic capacitance is an important requirement in USB 3.0 systems because excessive capacitive loading may degrade signal integrity and reduce high-speed communication reliability. Therefore, the ESD protection device is placed close to the USB Type-C connector in order to suppress transient events before they propagate into the high-speed switching circuitry.

In addition to data-line protection, the VBUS power line is protected using the PESD5V0L1UL ESD diode. Since the VBUS line is directly exposed through the USB Type-C connector, voltage transients generated during cable hot-plugging or electrostatic discharge may damage downstream power-routing circuitry. The PESD5V0L1UL diode provides transient voltage suppression for the VBUS path and improves overall system protection reliability.

One of the major challenges associated with USB Type-C interfaces is reversible connector orientation. In USB Type-C systems, the connector may operate in either normal orientation or flipped orientation. Due to this reversible behavior, the USB connector internally contains duplicate high-speed TX/RX lane assignments, but only one set of lanes becomes active depending on cable orientation. As a result, the routing architecture must dynamically determine which TX/RX lane set should be connected into the active USB communication path.

To address this requirement, the HD3SS3220 orientation detection IC is integrated into each USB Type-C input channel [1]. The HD3SS3220 monitors the CC1 and CC2 signals to determine connector orientation and automatically switches the corresponding high-speed TX/RX differential pairs internally. This implementation simplifies the downstream routing architecture because the orientation-handling and lane-selection operations are performed within a single integrated device. Consequently, the output of the HD3SS3220 provides a consistent high-speed signal interface independent of connector insertion orientation.

To improve supply stability and reduce high-frequency noise near the switching ICs, decoupling capacitors of 1 μF and 0.1 μF are connected in parallel at the supply pins of the high-speed switching devices. Multiple capacitors are used because capacitors of different values provide effective filtering characteristics across different frequency ranges. The 0.1 μF capacitor provides fast transient response and suppresses high-frequency switching noise generated during high-speed digital operation. However, smaller-value capacitors generally exhibit higher effective series resistance (ESR) and limited charge-storage capability, which may result in voltage fluctuations

during transient current demands. To reduce the overall effective ESR and provide improved local energy storage, a 1 μF capacitor is connected in parallel with the 0.1 μF capacitor. The combined decoupling network improves supply stability, reduces voltage ripple, and maintains reliable operation of the high-speed switching circuitry.

C. User Control and Power Regulation Architecture

User-controlled channel selection is implemented using the A30403RNCQ rotary switch [6]. The rotary switch provides a simple and reliable hardware-based channel-selection mechanism without requiring microcontroller-based control logic or firmware implementation. This approach simplifies the overall system architecture and reduces software dependency during USB path selection.

Different combinations of SEL_A and SEL_B correspond to different USB Type-C input channels. The SEL control signals are generated using a hardware-based 3P4T rotary switch configuration. The rotary switch selectively connects the SEL_A and SEL_B output lines to either 3.3 V logic HIGH or GND logic LOW depending on the selected switch position. By assigning different voltage combinations across the rotary-switch throws, unique binary control states are generated for each USB Type-C input channel.

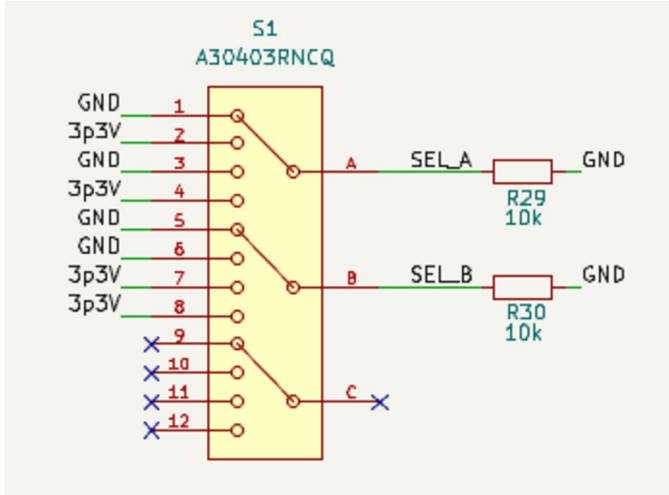


Fig. 2. Rotary-switch based generation of SEL_A and SEL_B control signals.

Pull-down resistors are connected to the SEL_A and SEL_B control lines to ensure stable logic states and prevent floating control signals during rotary-switch transitions. Floating logic signals may cause unintended switching behavior or temporary routing instability during user selection. Therefore, the pull-down network improves control reliability and ensures deterministic multiplexer operation.

The HD3SS3220 orientation controller, HD3SS6126 high-speed data multiplexer, and TMUX1574 CC multiplexer require a regulated 3.3 V power supply for proper operation. To generate this supply voltage, an external 5 V input source is connected to the TPS63001 DC-DC converter IC. The

TPS63001 generates a stable regulated 3.3 V output which is distributed throughout the switching architecture.

D. High-Speed Data Routing Architecture

The high-speed data routing block transfers USB 3.0 SuperSpeed differential signals between the selected USB Type-C input port and the USB Type-C output port. Since USB 3.0 operates at multi-gigabit data rates, signal integrity is an important design consideration.

Each USB Type-C connector supports reversible plug orientation, causing the active TX/RX lanes to change depending on cable insertion direction. To handle this orientation-dependent routing, an HD3SS3220 orientation-detection IC is integrated into every USB Type-C input channel. The HD3SS3220 monitors the CC1 and CC2 signals and internally maps the active SuperSpeed lanes into a fixed output configuration.

After orientation correction, the differential pairs are routed through a centralized HD3SS6126-based multiplexer tree architecture. Three HD3SS6126 high-speed multiplexers are used to implement four-input to single-output USB 3.0 routing [2]. The SEL_A and SEL_B control signals generated from the rotary-switch control block determine the active routing path.

AC-coupling capacitors are inserted along the SuperSpeed differential paths before entering the HD3SS6126 multiplexers. The USB 2.0 D+ and D− signals are also routed through the switching architecture to maintain device initialization and backward compatibility.

Differential TX/RX pairs are routed symmetrically with controlled spacing and minimal length mismatch to preserve signal integrity within the USB 3.0 communication channel.

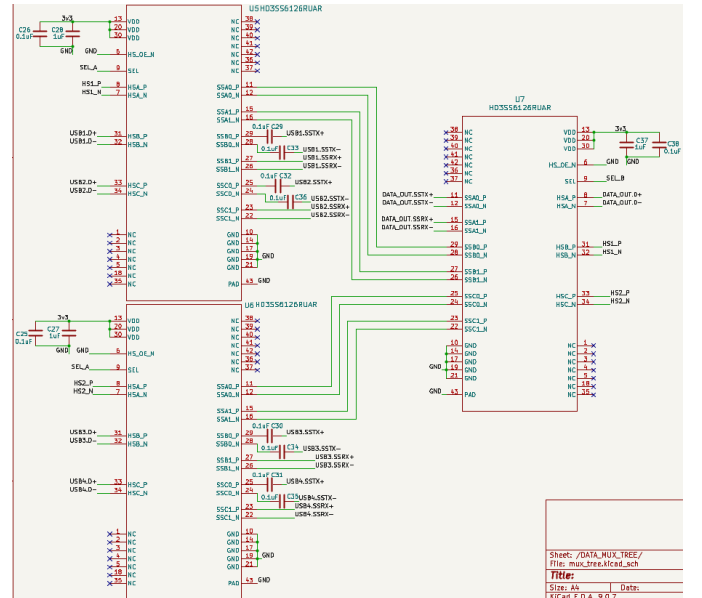


Fig. 3. Data Mux architecture.

E. Power Routing Architecture

The proposed architecture uses three TPS2121 power multiplexer ICs arranged in a hierarchical switching structure to

implement four-input to single-output VBUS routing. Two TPS2121 devices perform first-stage channel selection, while the third TPS2121 performs final output selection. The SEL_A and SEL_B control signals generated from the rotary-switch control block determine the active VBUS path.

The TPS2121 device was selected because it integrates seamless power switchover, reverse-current blocking, current limiting, and thermal protection within a single IC [3]. The device also supports fast switching operation with approximately 5 μ s switchover time, making it suitable for stable USB Type-C VBUS routing applications. A 22.1 kohm resistor is connected to the ILIM pin of each TPS2121 device to configure the output current limit. According to the TPS2121 datasheet, this resistor value sets the current limit to approximately 4.5A, providing sufficient current capability for USB peripheral and charging applications while protecting the power-routing path during overload conditions.

0.1 μ F decoupling capacitors are placed near the VBUS inputs to suppress transient voltage fluctuations generated during hot-plugging and power switching events. Wider PCB traces are also used in the VBUS routing path to reduce resistive losses and support higher load-current handling capability.

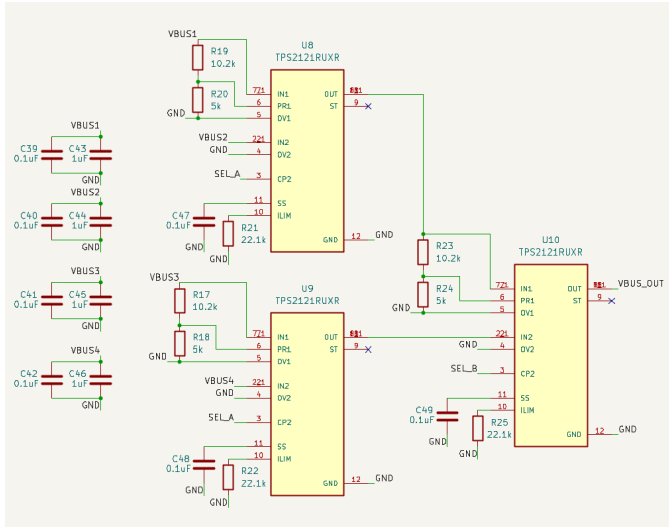


Fig. 4. Power Mux architecture.

F. CC Routing Architecture

Two TMUX1574 devices perform first-stage CC channel selection, while the third device performs final routing of the selected CC1 and CC2 signals to the output USB Type-C connector. The SEL_A and SEL_B control signals generated from the rotary-switch control block are used to synchronize CC routing with the data and VBUS switching paths.

The TMUX1574 device was selected due to its low ON resistance, low channel capacitance, and high bandwidth characteristics, [4] which are important for maintaining reliable USB Type-C CC communication. The device also supports bidirectional signal routing and powered-off protection, pre-

venting unintended back-powering conditions during switching operation.

0.1 μ F and 1 μ F decoupling capacitors are placed near the VDD pins of each TMUX1574 device to reduce supply noise and improve switching stability. Since CC lines are responsible for cable orientation detection and USB Type-C connection management, stable signal integrity is necessary for reliable device detection and communication establishment.

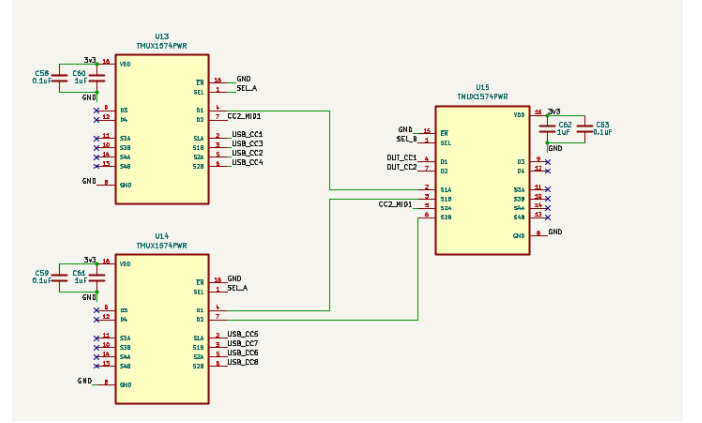


Fig. 5. CC Mux architecture.

III. SYSTEM FEATURES AND APPLICATION SUPPORT

The proposed USB Type-C switching architecture supports simultaneous routing of USB 3.0 high-speed data signals, VBUS power, and CC communication signals between four USB Type-C input ports and a single output port. The architecture provides transparent USB connectivity by ensuring synchronized switching of all communication paths using common SEL_A and SEL_B control signals.

The implemented design supports USB peripheral applications including keyboards, mice, USB storage devices, printers, webcams, micro-controller development boards, and standard USB hubs. Since the architecture maintains transparent CC routing between endpoint devices, USB Type-C connection detection and Power Delivery (PD) negotiation are handled directly by the connected devices without requiring dedicated PD controller implementation inside the switching system. Charging applications are also supported through the VBUS power-routing architecture.

The proposed system supports USB-based video transmission such as webcam streaming through the USB 3.0 high-speed data path. However, DisplayPort Alternate Mode and HDMI Alternate Mode operation are not guaranteed because the architecture does not implement complete USB Type-C alternate-mode handling. In addition, the use of a compact two-layer PCB introduces signal-integrity limitations at very high data rates and may affect performance in high-bandwidth video applications.

IV. RESULTS AND PCB IMPLEMENTATION

The proposed USB Type-C multi-port switching architecture was implemented as a compact two-layer PCB with an overall

board dimension of 80 mm × 80 mm. Fig. 6 shows the final 3D PCB layout of the implemented hardware design. The PCB integrates the USB Type-C input connectors, HD3SS3220 orientation-detection ICs, HD3SS6126 high-speed data multiplexers, TPS2121 power multiplexers, TMUX1574 CC multiplexers, DC-DC converter circuitry, and rotary-switch control mechanism within a compact layout structure.

Special PCB layout considerations were required due to the presence of USB 3.0 high-speed differential routing paths. Differential TX/RX pairs were routed symmetrically with controlled spacing and minimal length mismatch to maintain signal integrity throughout the communication path. Compact component placement was also maintained to reduce parasitic routing effects and minimize excessive trace discontinuities.

One of the major layout challenges was the mismatch between the TX/RX pin arrangement of the USB Type-C connector and the HD3SS3220 orientation-detection IC. This introduced routing complexity while maintaining differential-pair symmetry within the limited routing space available in a two-layer PCB implementation. Additional challenges were associated with simultaneous routing of multiple CC paths, VBUS power lines, and high-speed differential signals within the compact board area. Despite these routing constraints, the proposed architecture demonstrates a practical implementation approach for USB Type-C peripheral sharing and power-routing applications.

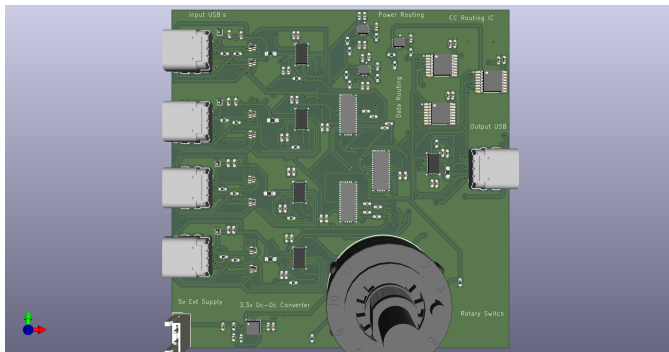


Fig. 6. 3D PCB Layout.

V. CONCLUSION

This paper presented a USB Type-C multi-port switching architecture capable of routing USB 3.0 data, VBUS power, and CC communication signals between four USB Type-C input ports and a single output port. The proposed architecture demonstrates transparent USB Type-C switching using synchronized data, power, and CC routing mechanisms. The implemented design provides a practical solution for USB peripheral sharing and power-routing applications using a compact two-layer PCB architecture. Future work may include support for USB Type-C alternate modes and higher-layer protocol-aware switching architectures.

REFERENCES

- [1] Texas Instruments, "HD3SS3220 USB Type-C DRP Port Controller Datasheet." Available: <https://www.ti.com/product/HD3SS3220>
- [2] Texas Instruments, "HD3SS6126 SuperSpeed USB 3.0 Multiplexer Datasheet." Available: <https://www.ti.com/product/HD3SS6126>
- [3] Texas Instruments, "TPS2121 Power Multiplexer Datasheet." Available: <https://www.ti.com/product/TPS2121>
- [4] Texas Instruments, "TMUX1574 Precision Analog Switch Datasheet." Available: <https://www.ti.com/product/TMUX1574>
- [5] CNC Tech, "C-ARA1-AK511-1 USB Type-C Receptacle Connector." Available: <https://www.digikey.in/en/products/detail/cnc-tech/C-ARA1-AK511-1/25330270>
- [6] C&K, "A30403RNCQ Rotary Switch Datasheet." Available: <https://www.digikey.in/en/products/detail/c-k/A30403RNCQ/10270592>